

Radicals, Rational Exponents, Domains, and Approximate Solutions

Answer Key

Algebra 2 Radical Expressions and Equations

Quick Answers

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|----------------|----------------|-----------------------|
| 1. 7 | 5. 9 | 9. 0.25 |
| 2. $5\sqrt{2}$ | 6. $9\sqrt{3}$ | 10. 4 |
| 3. 9 | 7. 3 | 11. $x^{\frac{5}{6}}$ |
| 4. 4.47 | 8. 4.24 | 12. 1, 2.45 |
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Common wrong answers

4. If they got 6.32: simplified $\sqrt{20}$ as $2\sqrt{10}$ by halving the radicand instead of factoring out a perfect square
8. If they got 3.16: added the radicands and wrote $\sqrt{10}$
9. If they got -4: made the answer negative instead of taking the reciprocal
12. If they got -2.45: kept the negative root even though it makes the right side of the equation negative
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1. Evaluate $49^{1/2}$.

Step 1. A fractional exponent names a root: the denominator 2 says square root, so no calculator is needed once the perfect square is recognised.

Step 2. $49^{1/2} = \sqrt{49} = 7$, the principal (non-negative) root.

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In words: the one-half power asks “what non-negative number squares to 49?”

2. Simplify $\sqrt{50}$.

Step 1. Simplifying means removing perfect-square factors, so factor the radicand with that goal: $50 = 25 \cdot 2$, and 25 is the largest perfect square dividing 50.

Step 2. $\sqrt{50} = \sqrt{25 \cdot 2} = 5\sqrt{2}$.

$5\sqrt{2}$

Check: $(5\sqrt{2})^2 = 25 \cdot 2 = 50$.

3. Solve $\sqrt{x+7} = 4$.

Step 1. State the domain before touching the equation: the radicand must be non-negative, so $x + 7 \geq 0$, i.e. $x \geq -7$. The right side is 4, which is positive, so squaring will not manufacture a sign error here.

Step 2. Squaring gives $x + 7 = 16$, so $x = 9$.

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Check: $\sqrt{9+7} = \sqrt{16} = 4$. True, and $9 \geq -7$, so the solution is in the domain.

4. Approximate $\sqrt{20}$.

Step 1. Simplify before approximating, so the decimal comes from a small familiar root rather than from a large one: $20 = 4 \cdot 5$, so $\sqrt{20} = 2\sqrt{5}$.

Step 2. $\sqrt{5} \approx 2.2361$, so $2\sqrt{5} \approx 4.4721$, which rounds to 4.47.

4.47

The trap: halving the radicand to get $2\sqrt{10} \approx 6.32$ is not a legal move — only a perfect-square *factor* comes out, and it comes out as its root.

5. Evaluate $27^{2/3}$.

Step 1. Denominator first: the 3 says cube root, and taking the root before the power keeps the arithmetic small.

Step 2. $\sqrt[3]{27} = 3$, then the numerator squares it: $3^2 = 9$.

9

Either order works — $\sqrt[3]{27^2} = \sqrt[3]{729} = 9$ — but the root-first order avoids the three-digit number.

6. Combine $3\sqrt{12} + \sqrt{27}$.

Step 1. Radicals are like terms only when the radicands agree *after* simplifying, so simplify both before comparing them.

Step 2. $\sqrt{12} = 2\sqrt{3}$, so $3\sqrt{12} = 6\sqrt{3}$; and $\sqrt{27} = 3\sqrt{3}$.

Step 3. Both are multiples of $\sqrt{3}$, so they add: $6\sqrt{3} + 3\sqrt{3} = 9\sqrt{3}$.

9 $\sqrt{3}$

Why the first look deceives: 12 and 27 are different radicands, so the terms appear unlike until each is simplified.

7. Solve $\sqrt{2x+3} = x$.

Step 1. The left side is a principal square root and therefore never negative, so any solution must have $x \geq 0$; note that restriction before squaring, because squaring destroys sign information.

Step 2. Squaring: $2x + 3 = x^2$, so $x^2 - 2x - 3 = 0$, which factors as $(x - 3)(x + 1) = 0$, giving candidates $x = 3$ and $x = -1$.

Step 3. Test both. $x = 3$: $\sqrt{9} = 3$. True. $x = -1$: $\sqrt{1} = 1$, but the right side is -1 . False.

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Why -1 fails: it satisfies the *squared* equation, since squaring turns $1 = -1$ into $1 = 1$. The original equation asks for a non-negative left side to equal a negative right side, which is impossible.

8. Approximate $\sqrt{2} + \sqrt{8}$.

Step 1. Combine before approximating: rounding twice and then adding compounds the error, and the terms turn out to be like terms.

Step 2. $\sqrt{8} = 2\sqrt{2}$, so the sum is $\sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$.

Step 3. $\sqrt{2} \approx 1.4142$, so $3\sqrt{2} \approx 4.2426$, which rounds to 4.24.

4.24

Why it is not $\sqrt{10}$: $\sqrt{10} \approx 3.16$, which is not the same number. Radicands add only under multiplication, never under addition — $\sqrt{a} + \sqrt{b} \neq \sqrt{a+b}$.

9. Evaluate $16^{-1/2}$.

Step 1. Take the two pieces of the exponent one at a time. The minus sign means reciprocal, not a negative value, so the answer will be positive; the $\frac{1}{2}$ means square root.

Step 2. $16^{-1/2} = \frac{1}{16^{1/2}} = \frac{1}{\sqrt{16}} = \frac{1}{4}$.

 $\frac{1}{4} = 0.25$

The trap: answering -4 reads the minus as a sign on the result. A negative exponent flips the base; it cannot make a positive base produce a negative value.

10. Solve $\sqrt{x+5} = x-1$.

Step 1. The left side is non-negative, so the right side must be too: $x-1 \geq 0$, i.e. $x \geq 1$. That single inequality will settle the extraneous candidate in advance.

Step 2. Squaring: $x+5 = x^2-2x+1$, so $x^2-3x-4 = 0$, which factors as $(x-4)(x+1) = 0$: candidates $x = 4$ and $x = -1$.

Step 3. $x = 4$: $\sqrt{9} = 3$ and $4-1 = 3$. True. $x = -1$: $\sqrt{4} = 2$ but $-1-1 = -2$. False, and it also violates $x \geq 1$. 4

Note: $x = -1$ is in the domain of the radical ($-1+5 = 4 \geq 0$). Being in the domain is necessary but not sufficient — the sign condition on the other side is the extra requirement.

11. Write $x^{1/2} \cdot x^{1/3}$ as a single power.

Step 1. The bases are identical, so the product rule applies and the exponents add — there is no need to convert to radicals at all.

Step 2. $\frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$, so the product is $x^{5/6}$. $x^{5/6}$

In radical form: $\sqrt[6]{x^5}$ — the same object, which is exactly what rational exponents are for.

12. Challenge: two near-identical radical equations.

Step 1. Both have a principal square root on the left, so in both the right side must be non-negative; that condition, not the algebra, is what disqualifies a candidate later.

Step 2. Equation (i), $\sqrt{2x+7} = x+2$. Squaring: $2x+7 = x^2+4x+4$, so $x^2+2x-3 = 0$, factoring as $(x+3)(x-1) = 0$: candidates $x = -3$ and $x = 1$. Test: $x = 1$ gives $\sqrt{9} = 3$ and $1+2 = 3$, true; $x = -3$ gives $\sqrt{1} = 1$ but $-3+2 = -1$, false. 1

Step 3. Equation (ii), $\sqrt{2x+7} = x+1$. Squaring: $2x+7 = x^2+2x+1$, so $x^2 = 6$ and the candidates are $x = \sqrt{6}$ and $x = -\sqrt{6}$. The negative one makes the right side $-\sqrt{6}+1 < 0$, so it is extraneous. The survivor is $x = \sqrt{6} \approx 2.4495$, which rounds to 2.45. 2.45

Part (c): squaring is not reversible. It maps $A = B$ and $A = -B$ to the same equation, so the squared equation carries solutions of the sign-flipped problem as well. The substitution check is what removes them.